

# CHAE YOUNG SIM

📍 Seoul    ✉️ simcy1024@korea.ac.kr    ☎️ +82-10-8728-9314

## RESEARCH INTERESTS

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Computer Architecture, Dynamic Thermal Management, Thermal Modeling

## EDUCATION

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### Korea University

Mar 2025 – Feb 2030 (Expected)

*Ph.D. in Computer Science and Engineering*

*Seoul, Korea*

- Advised by Professor Sung Woo Chung
- GPA: 4.43 / 4.5
- Coursework: Computer Architecture and Systems, High-Performance Computing Systems, Semiconductor Materials and Devices, etc.

### Korea University

Mar 2021 – Feb 2024

*B.S. in Computer Science and Engineering*

*Seoul, Korea*

- GPA: 4.04 / 4.5
- Coursework: Computer Architecture and Systems, Operating Systems, Computer Network, Machine Learning, Deep Learning, etc.

## PUBLICATIONS

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Jae Yoon Lee\*, **Chae Young Sim\***, Seung Hun Choi, and Sung Woo Chung, “**Thermal Challenges and Opportunities for Off-the-shelf 3D-stacked CPUs**,” *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2025. \*These authors contributed equally to this work.

(Domestic) **Chae Young Sim**, Jae Yoon Lee, and Sung Woo Chung, “**Performance Comparison of Heterogeneous Cores in Mobile APs under Thermal Constraints**,” *Korea Computer Congress (KCC)*, 2025.

Jihyun Kim, Chaeyeon Lee, Jisoo Song, **Chae Young Sim**, and Seongbin Park, “**Teaching an Elective Course about Quantum Computing**,” *International Conference on Informatics in Schools: Situation, Evolution, and Perspectives (ISSEP)*, 2023.

## EXPERIENCE

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### Research Assistant

Mar 2025 – Current

*SoC & Microprocessor Research Lab. (Advisor: Prof. Sung Woo Chung)*

*Seoul, Korea*

- Conducted experiments on Samsung Exynos 2600 development board (Thetis) to build a temperature-aware performance and power prediction tool. (Supported by Samsung Electronics’ System LSI)
- Designed a dynamic thermal and power management technique for heterogeneous core systems (e.g., CPU, GPU, and NPU), exploiting instruction complexity in software-defined robotics (SDR) workloads. (Supported by IITP)
- Developed OS-level thermal-aware task scheduling techniques for high-performance CPUs, leveraging floorplan and adaptive voltage scaling (AVS). (Supported by IITP)
- Conducted experiments on Samsung Exynos 2600 development board (Thetis) to build a temperature-aware performance and power prediction tool. (Supported by Samsung Electronics’ System LSI)

### Undergraduate Researcher

Sep 2024 – Feb 2025

*SoC & Microprocessor Research Lab. (Advisor: Prof. Sung Woo Chung)*

*Seoul, Korea*

- Analyzed performance differences between big core and middle core in mobile APs under thermal constraints, highlighting the need for thermal-aware SoC design especially in big core. (Supported by IITP)

### Undergraduate Researcher

*Machine Learning & Vision Lab. (Advisor: Prof. Hyunwoo J. Kim)*

Jul 2023 – Jan 2024

*Seoul, Korea*

- Studied artificial general intelligence (AGI), generative models, diffusion.

### AI Software Developer

*Doctorstech, under NICE*

Jan 2023 – Feb 2023

*Seoul, Korea*

- Developed a beauty trend identifying service using Python.
- Helped develop a game device, an acoustically responsive vest.

### Undergraduate Researcher

*Algorithm Lab. (Advisor: Prof. Seongbin Park)*

Jul 2021 – May 2023

- Studied theory of computation, quantum computing algorithms.
- Inspected the upcoming book, *Algorithm And Hyperlinks*.

## PATENTS

(Domestic) Sung Woo Chung, Jae Yoon Lee, and **Chae Young Sim**, “**Method and apparatus for heterogeneous-core migration**,” *Korea Application Number: 10-2026-0015657*, 2026.

## PROFESSIONAL EXPERIENCE

### Reviewer

*IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*

2025 - 2026

### Teaching Assistant

*Computer Architecture (Instructor: Prof. Sung Woo Chung)*

Fall 2025

*COSE222*

- Undergraduate course (50+ students).
- Delivered six 1.25 hour lectures covering Verilog HDL fundamentals, logic synthesis, and functional/timing simulation using an Altera DE2 FPGA board.
- Designed Verilog HDL programming assignments, including combinational/sequential circuits, a RISC-V single-cycle processor, and RISC-V pipeline processor.

## AWARDS AND HONORS

### Best Teamwork

*Google Datacenter Hardware Hackathon*

Aug 2024

*Seoul, Korea*

## SKILLS AND TECHNIQUES

### Programming

*Python, PyTorch, C, C++, OCaml, PHP, ARM Assembly, RISC-V Assembly*

### Languages

*Native Korean, Conversational English*

## REFERENCES

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**Sung Woo Chung**

Professor

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